

AI Usage Log

EECS 348 - Lab 4
Student: Conor Rothe
AI Tool: ChatGPT

This log summarizes the ChatGPT assistance used while completing and preparing EECS 348 Lab 4. It records the main questions asked, troubleshooting performed, commands discussed, and final file organization.

1. Accessing the Cycle server and locating Lab 4

User: User asked how to return to the KU Cycle server and navigate to the Lab 4 directory.

ChatGPT: ChatGPT guided the user through SSH/Cycle access and directory navigation. The working directory was checked with commands such as `pwd`, `ls`, and `cd ~/EECS348Lab4`.

2. Compiling with the Makefile

User: User attempted to run `make`. The linker returned: `cannot open output file football_scores: Permission denied`.

ChatGPT: ChatGPT identified that the output executable could not be created because of permissions and helped inspect the directory and file permissions using `ls -l`. After permissions were corrected, `make` successfully compiled both programs.

3. Testing Task 1 - Football Scores

User: User asked what to do after compilation and how to run the first task.

ChatGPT: ChatGPT instructed the user to run the football scoring executable and enter test scores. The program successfully displayed possible NFL scoring combinations and accepted 1 to stop.

4. Testing Task 2 - Temperature Conversion

User: User asked what temperature to enter and what to do next.

ChatGPT: ChatGPT suggested a simple test value and walked through the temperature program's prompts. The program successfully converted temperatures and displayed a temperature category and weather advisory.

5. Copying files from Cycle to Windows

User: User needed the compiled/work files back on the laptop so they could be placed into the Git repository.

ChatGPT: ChatGPT provided `scp` commands to copy the files from `~/EECS348Lab4` on Cycle into the local EECS348 directory on Windows. The user verified the files with `dir`.

6. Organizing the GitHub repository

User: User asked whether to use GitHub Desktop or the website after initially creating the repository on the GitHub website.

ChatGPT: ChatGPT recommended using GitHub Desktop for the local workflow and helped locate the local repository folder in File Explorer.

7. Renaming the source files

User: User needed the source files renamed to Task 1 and Task 2.

ChatGPT: ChatGPT guided the user in renaming `football_scores.c` to `task1.c` and `temperature.c` to `task2.c` while preserving the `.c` extensions.

8. Updating the Makefile

User: After renaming the C source files, the Makefile still referenced the old `football_scores.c` and `temperature.c` names.

ChatGPT: ChatGPT helped update the Makefile so the targets were `task1` and `task2` and the source files were `task1.c` and `task2.c`. The clean target was also changed to remove `task1` and `task2`.

9. Rebuilding on Cycle

User: User needed to demonstrate that the renamed programs could still compile and run on the Cycle server.

ChatGPT: ChatGPT walked through copying `task1.c`, `task2.c`, and the updated Makefile to Cycle, then rebuilding with `make clean` and `make all`.

10. Commands used for final Cycle test

ChatGPT provided/confirmed commands including:

```
cd ~/EECS348Lab4
make clean
make all
./task1
./task2
ls
```

The user demonstrated Task 1 with valid and invalid football-score inputs and Task 2 with a temperature conversion. Both renamed programs compiled and ran successfully.

11. Removing obsolete files

User: The instructor wanted the old `football_scores` and `temperature` files removed, and the user asked how to finish the submission.

ChatGPT: Explained that the obsolete old-name files should be removed so the final project uses the `task1/task2` naming. The final repository source setup should retain the Makefile, `task1.c`, and `task2.c`. Compiled executables can be recreated with `make` when needed.

Final Lab 4 file structure

```
EECS348Lab4/
  Makefile
  task1.c
  task2.c
```

Summary of AI usage: ChatGPT was used for command-line guidance, troubleshooting compilation and permission errors, testing suggestions, file transfer instructions, file-renaming guidance, Makefile updates, Cycle compilation/testing, and Git/GitHub repository organization.